

Q22. Principle of Optimality vs Greedy Choice Property (Conceptual Comparison)

Aim:

To compare the Principle of Optimality used in Dynamic Programming and the Greedy Choice Property used in Greedy Algorithms for solving resource allocation problems.

Algorithm / Steps:

Dynamic Programming (Principle of Optimality)

1. Divide the problem into smaller subproblems.
2. Solve each subproblem optimally.
3. Store intermediate results.
4. Build the final solution from optimal subproblem solutions.

Greedy Method (Greedy Choice Property)

1. Select the best immediate choice.
2. Never reconsider previous choices.
3. Repeat until the solution is complete.

Comparison Table:

Feature	Dynamic Programming	Greedy Algorithm
Principle	Principle of Optimality	Greedy Choice Property
Decision Making	Considers all possibilities	Chooses locally best option
Storage	Uses memoization/table	No extra storage
Optimal Solution	Always (if DP applicable)	Only if greedy property holds
Speed	Usually slower	Usually faster
Example	0/1 Knapsack	Fractional Knapsack

Conditions Where Both Hold:

- Minimum Spanning Tree
- Dijkstra's Algorithm (non-negative weights)
- Huffman Coding

Limitations:

When Greedy Fails

Example: 0/1 Knapsack

Items:

Weight	Value
10	60
20	100
30	120

Capacity = 50

Greedy may not give the optimal answer, while DP gives the optimal value of 220.

When DP is Costly

- Requires more memory.
- Higher computation time.

Time Complexity:

Method	Time Complexity	Space Complexity
Dynamic Programming	$O(n^2)$ (typical)	$O(n^2)$
Greedy	$O(n \log n)$	$O(1)$

Result:

Thus, Dynamic Programming guarantees optimal solutions through the Principle of Optimality, whereas Greedy Algorithms rely on locally optimal decisions using the Greedy Choice Property. The choice depends on whether the problem satisfies the required properties.